



SMALL FOOTPRINT CORING RIG

➤ Executive Summary

➤ Temperature Gradient Applications

➤ Potential Provider Capabilities

➤ Helicopter Coring Rig Mobilization Sequences

➤ Key Considerations

➤ Findings

➤ Roughly Commercial View

➤ Conclusions

➤ Follow-up Research



EXECUTIVE SUMMARY

TYPICAL WELL SCHEMATIC

EXECUTIVE SUMMARY



THIS STUDY IS TO INVESTIGATE THE FEASIBILITY OF UTILIZING A SMALL FOOTPRINT / HELICOPTER DEPLOYED CORING RIG TO DRILL TEMPERATURE GRADIENT WELLS UP TO +/- 800 M TO SUPPORT GEOLOGICAL MODELLING ON POTENTIAL GEOTHERMAL SITES.



THE TEAM COMPRISING RESOURCES, DRILLING AND GEOTHERMAL PERSONNEL HAVE SOURCED POTENTIAL SERVICE AND RIG PROVIDERS WITH EXPERIENCE IN OPERATION OF HELICOPTER DEPLOYED CORING RIG IN INDONESIA AND WORLDWIDE.

TYPICAL WELL
SCHEMATIC

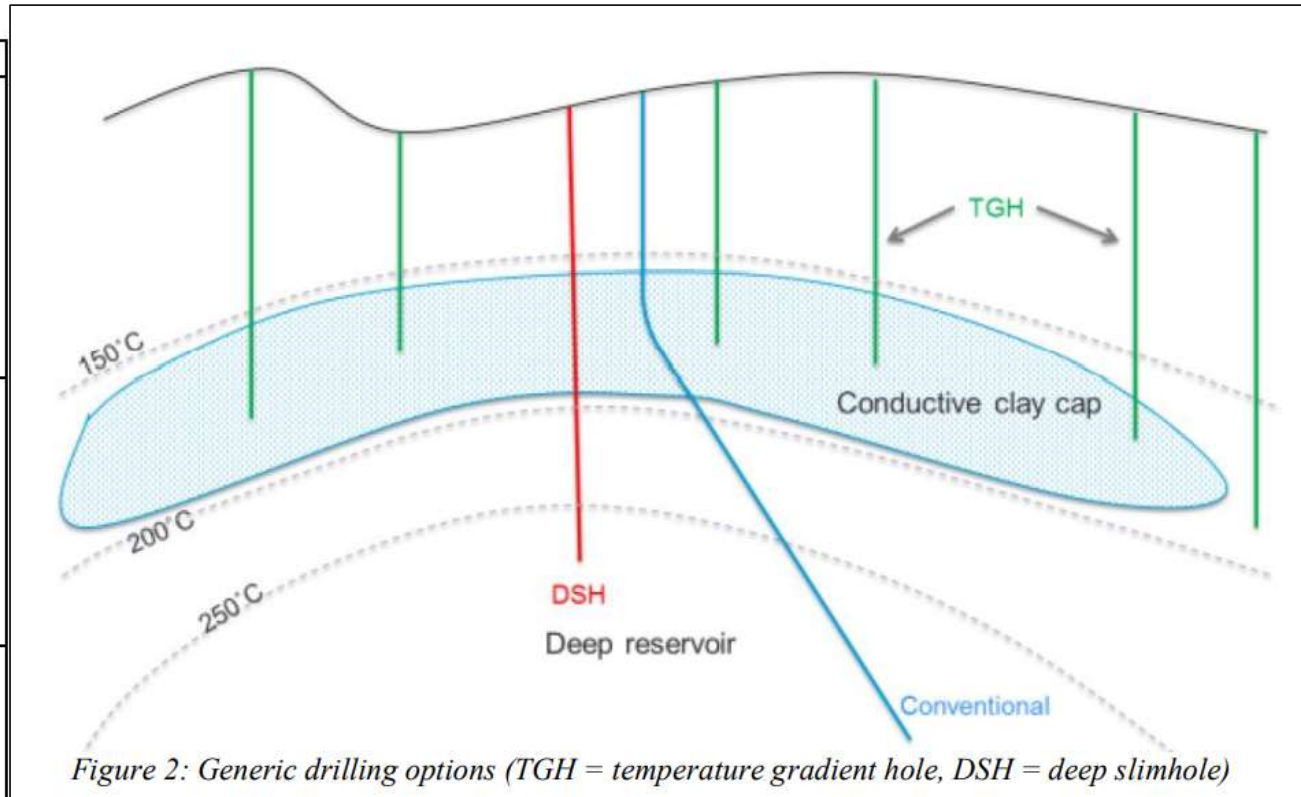
TEMPERATURE GRADIENT WELL

TEMPERATURE GRADIENT WELL APPLICATION

Table 1: Generic Well Definitions and Applications

Type	Variant	Description
Slimhole Well	Temperature Gradient	Well completed above the level of the productive reservoir to establish presence of clay conductor, confirm temperature, and validate conceptual model. ~200 – 800 mVD.
	Deep Slimhole (2 3/4" – 7" PC)	Drilled into the reservoir to confirm commercial temperature (primary objective) and can test productivity (secondary objective). ~500 – 2,000 mVD.
Conventional Well	Standard (9 5/8" PC)	Wells that penetrate substantial thickness of the reservoir, enable comprehensive testing of resource productivity, and may be used for production or injection in a development. ~2,000 – 3,000 mVD.
	Large (13 3/8" PC)	

Notes: PC = production casing, VD = vertical depth

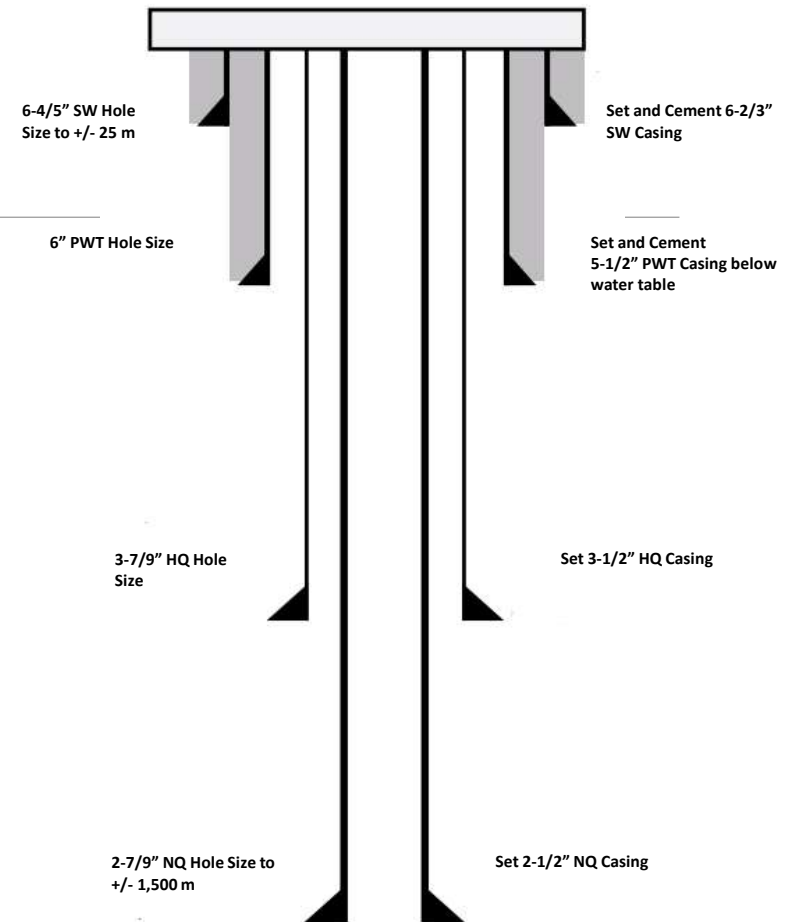


USE OF DEEP SLIMHOLE DRILLING FOR GEOTHERMAL EXPLORATION

K.M. Mackenzie, G.N.H. Ussher, R.B. Libbey, P.F. Quinlivan, J.U. Dacanay and I. Bogie

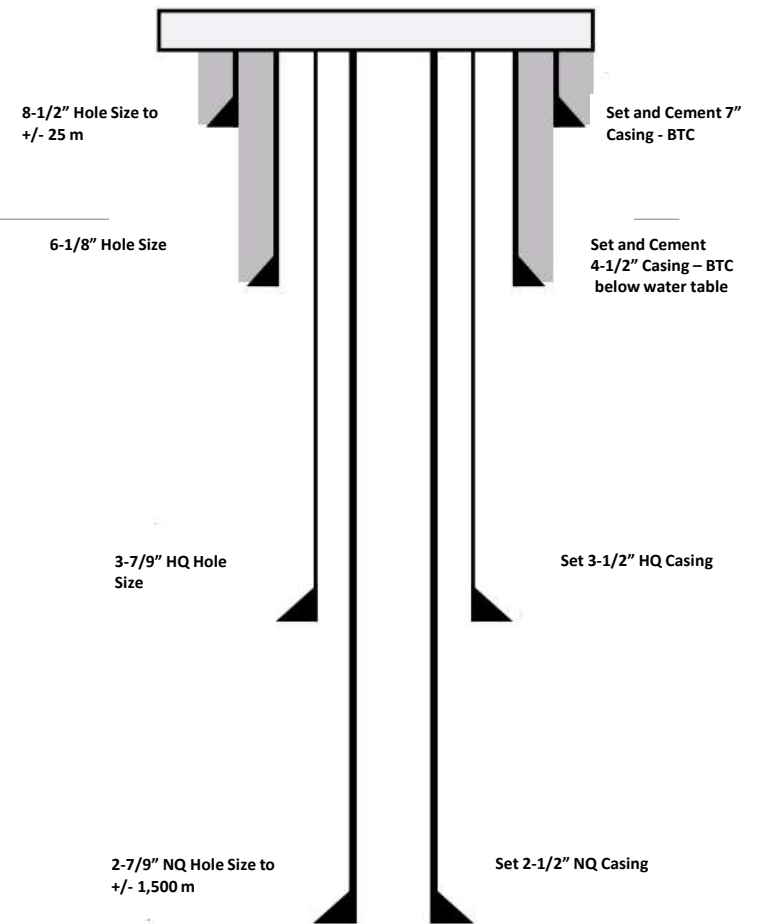
SLIM HOLE WELL CATEGORY

Slim hole Type	Well Depth	Rig Type
Temperature Gradient Well	- Limited to several hundreds' meters below water table	Helicopter Coring Rig
Deep Slimhole (2-3/4" – 7" PC)		Hybrid Rig. Ex: predator-250 ??



SLIM HOLE WELL CATEGORY

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Temperature Gradient Well	- Limited to several hundreds' meters below water table	Helicopter Coring Rig
Deep Slimhole (2-3/4" – 7" PC)		Hybrid Rig. Ex: predator-250





POTENTIAL PROVIDERS WITH CAPABILITIES

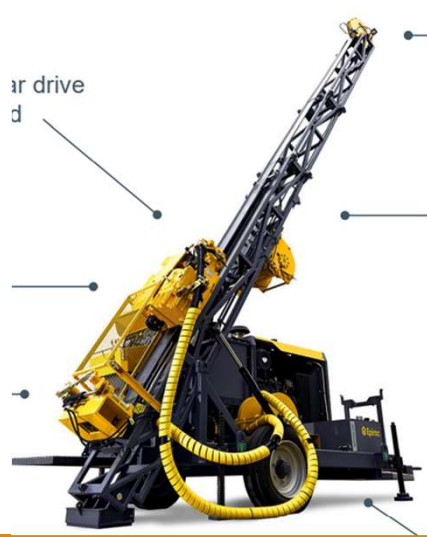
Potential Provider	OEM	Selling Rig	Provide Services	Similar Experience	Lead Time
Boart Longyear	YES	YES	YES	YES	3-4 Months
Epiroc	YES	YES	NO	NO	4-5 Months
Major Drilling	NO	NO	YES	YES	3 Months
Indo Drill	NO	YES	YES	YES	3 Months - new 4 Months - refurbishment
PT. CIME	NO	YES	YES	NO	3 Months
Sandvik	Bought by DrillMan (Australia) and no operation in Indonesia				
Mincon	YES	YES	NO	NO	No Info

POTENTIAL SERVICE PROVIDERS AND CAPABILITIES

Rig Candidate	Loads Number and Weight			Coring Depth Capabilities				Rig Properties				
	Number of Load	Maximum Weight	Total Weight	PQ	HQ	NQ	BQ	Engine	Pull Back	Pull Down	BOP Proposal	Rig Base
LF-90	12-14 Loads	670 Kg	5.4 Tons	476 m	722 m	1,064 m	1,385 m	220 HP	37.7 Klbs	19.8 Klbs	Annular Bag	Crawler with Jack
Boyle C-5	8 Loads	900 Kg	5 Tons	300 m	500 m	800 m	1,000 m	220 HP	24 Klbs	13.3 Klbs	No Proposal	Skid or Crawler
SCD-1,800	16 Loads	750 Kg	9.2 Tons	700 m	1,200 m	1,700 m	2,000 m	350 HP	41.5 Klbs	13.6 Klbs	Annular Bag	Skid
ID 2,000	20 Loads	705 Kg	9.6 Tons	800 m	1,500 m	2,000 m	3,000 m	350 HP	44 Klbs	13.6 Klbs	Annular Bag	Skid
CD 2,000	16 Loads	750 Kg	9.2 Tons	700 m	1,050 m	1,570 m	-	220 HP	30 Klbs	15 Klbs	API standard BOP	Crawler with Jack



LF-90



Boyle-C5



SCD - 1,800

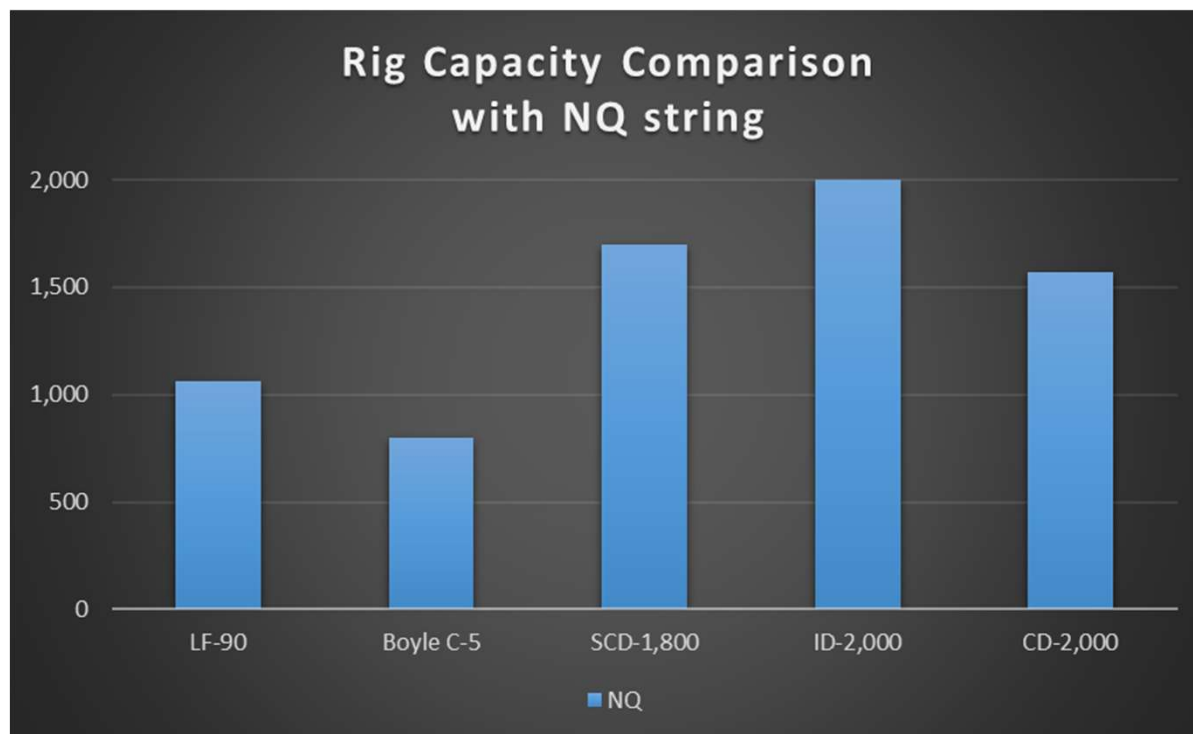


SCD - 2,000



Copyright

CD - 2,000



POTENTIAL SERVICE PROVIDERS AND CAPABILITIES

MOBILIZATION SEQUENCES

SMALL FOOTPRINT /
HELICOPTER CORING
RIG



SEA FREIGHT

Town to Sea Port



LAND TRANSPORT

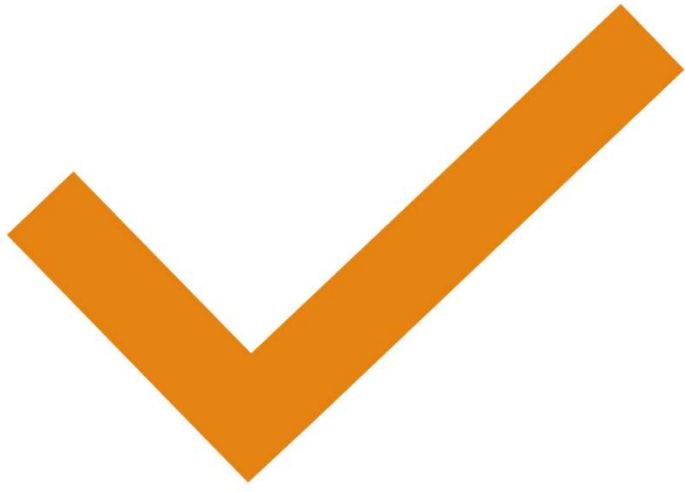
Sea Port to Staging Area



HELICOPTER TRANSPORT

Staging Area to Well Pad

OPERATIONAL SEQUENCE HELICOPTER RIG



KEY CONSIDERATIONS



Things to Consider	Remark
SKPI for rig	Permit of rig utilization
Yard rental	May be in service provider yard or the nearest project cluster
Maintenance during standby	It is recommended to have the rig manufacturing that can provide maintenance service to Ormat
Crew to run the rig during operation	It is recommended to have the rig manufacturing that can provide drilling service and design to Ormat

BUY,
MAINTAIN
AND RUN
THE CORING
RIG

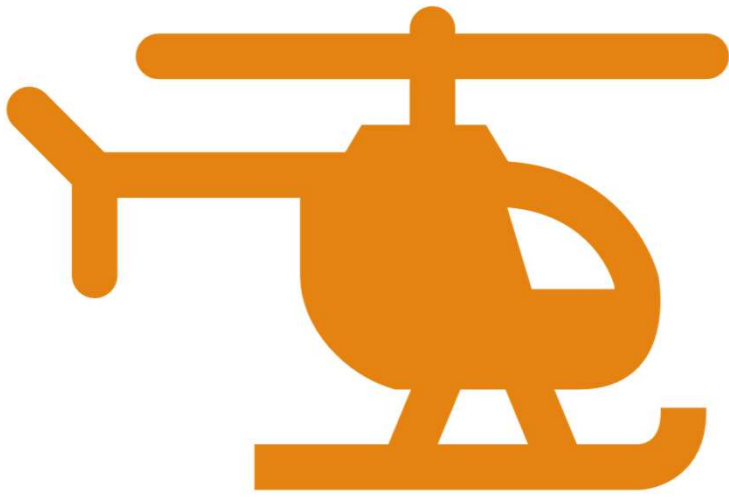
Things to Consider	Remark
Nearby Staging Area with accommodation	Maximum 1 hours from well coordinate
Helipad permit at staging area	Indonesia Air Transport Agency
Helipad construction permit	From local government agency
Heliport	Indonesia Air Transport Agency and Helicopter operation Manual
Things to Consider	Remark
Staging Area	Prepare the staging area and road access to staging area
	Staging Area maintenance and security system

SMALL FOOTPRINT WELL PAD

Things to Consider	Remark
Well pad Area	Build up by hand tools, to 30 m x 30 m
Crew Main Camps	Can be places at staging area or nearby village
Water source - Minimum water storage of 12 hours x 50 gpm -> 860 bbls (136,260 Lt)	Refers to NZS Standard (geothermal standard), Water storage should be able to accommodate a minimum of 12 hours of blind drilling without any Re-fill job
Land mobilization for small footprint rig	For the well pad that is possible to have a minimal civil infrastructure, land transport can be considered.
Well Abandonment Plan	Access to well monitoring, if the well is not P&A
Medical services	On site qualified medic and medical facilities and arrange medical evacuation services needs to be discussed
Cellar Construction	Cellar with 0.5 m – 1 m depth may be needed to accommodate the height of BOP
Communications	Two-way communication from staging area to well pad is needed.

Things to Consider	Remark
Well control system	Further study needed to design the wellhead system and BOP equipment
Well Design	Further study of well design needed in conjunction with BOP
Cementing design	Cementing job and slurry needs to be designed form small annulus clearance

WELL CONTROL AND WELL DESIGN



HELICOPTER SERVICES

AIR BALI HELICOPTER PROVIDER

Description	Detail
Helicopter Type	Bell 407
Load	600-750 kg
Parameters	Humidity, wind speed, elevation, temperature, timing - morning
Passengers	6 persons
Customers	Freeport, Valley, Anam Gold



Type	Detail
Staging Area	Prepare the staging area and road access to staging area
	Staging Area maintenance
	Staging Area Security
	Waste disposal – including cutting disposal
Licenses / Permit	Civil work license to staging area
	Permit for water supply
	Permit for Helipad construction from local government
	Permit to operate the Helipad from Indonesia Air Transportation
Staging Area Consideration	
Accommodation and facilities	Food, messing, Laundry, accommodation, camp facility, portable water
Communication	Two-way communication from staging area to well pad is needed.
	Staging area needs to be able to store related tangible items and other rig supporting equipment

HELICOPTER PREPARATION



HELICOPTER ROUGHLY COMMERCIAL CALCULATION

- Monthly fix cost @ +/- USD 110,000
- Usually, one year contract
- Hourly operation cost @ +/- USD 1,250
- Minimum monthly operation - 60 hours
- Maximum monthly operation – 120 hours
- Needs two Helicopter package beyond 120 hours utilization
- Fuel is excluded
- Maximum hours for pilot
 - 5 hours per day
 - 40 hours per week
 - 100 hours per month

Potential monthly exposure @ USD 185,000

- Monthly fix cost @ USD 110,000
- Monthly minimum rental @ USD 75,000



FINDINGS

FINDINGS

- There is no Helicopter coring rig operation for geothermal exploration currently being deployed in Indonesia.
- The potential service providers are currently providing the helicopter coring operation service to mineral mining companies.
- All of the potential services providers approached are capable of meeting the minimum requirements of this study.
- Only one of the vendors who responded offered a new rig, while the others offered refurbished rig. The new rig offer is limited to the rig sale only.
- All the refurbished rig providers offered crew and support services.
- Service providers contacted have proposed generic well designs. These may or not be suitable for Geothermal applications.
- Well control. Two of the responders offered non-API standard BOP system. One of the responders offered a conceptual API standard BOP system, however, this system has never been deployed.
- The Indonesian market for available Helicopter services, at this moment is limited. The most commonly used Helicopter is the Bell 407 with a maximum lift of 750 kg.
- All of the refurbished rigs have component parts weighing less than 750 kg. The new purchase rig would require a larger helicopter than the current market norm.
- Both new and refurbished rigs are available to the market USD 350,000 – 450,000 and 3-4 Months lead time.
- Based on a drilling campaign of 60 days for crew and coring rig estimate, 24 hours operation ~USD 420,000. This excludes the helicopter, drilling consumables, and import duty. This compares with estimated of ~USD 1,300,000 for 1,000 – 1,500 HP rig.

Decision Criteria		Yes	Maybe	No			
		3	2	1			
Parameters		Boart Longyear	EPIROC	IndoDrill	IndoDrill	PT. CIME	Remark
Capability of rig manufacturing and providing services to Ormat	Rig Candidate	FL-50	Boyle-C5	SCD-1,800	ID-2,000	CD-2,000	
	OEM	3	3	1	1	1	
	In Country	3	1	3	3	1	
	Provide Crew and Services	3	1	3	3	3	
	Lead Time - Less then 6 Months	3	3	3	3	3	
Experience of similar operation - Helicopter Coring Rig - Temperature Gradient Well	Similar Experience in Country	3	1	3	3	1	
Load Detail for Helicopter Transport	Number of Load	2	3	1	1	1	Less then 10 Loads -> 3 10-15 loads -> 2 More Then 15 Loads -> 1
	Maximum individual load weight	3	1	3	3	3	Less then 750 Kg -> 3 751-899 Kg -> 2 More Then 900 Kg -> 1
Capability to drill to proposed Depth : +/- 300 m below water level or total depth of 800 m with minimum NQ size - larger than the PT tools (> 1.75")	NQ	3	3	3	3	3	NQ to 800 m or greater - 3
API BOP System	BOP Proposal	1	1	1	1	3	API BOP System
Commercial proposal	USD Number * CIME is not coming with the commercial proposal	2	3	2	2	1	less then 350,000 USD -> 3 Between 350,000 - 450,000 USD -> 2 above 450,000 -> 1
Total Score		26	20	23	23	24	

DECISION CRITERIA



ROUGHLY COMMERCIAL VIEW

BOART LONG YEAR LF90

Section	Estimated Cost (USD)	No of Days	Total (USD)	Remarks
Refurbishment	400,000	-	400,000	
Mobilization	30,000	-	30,000	
Preparation	2,500 per shift	30	75,000	One shift per day
Drilling	3,500 per shift	40	280,000	10 m per shift
Item excluded: Fuel, mud and cementing materials, LIH / cemented pipe, coring road and downhole tools				

- § BLY have two LF-90 in country, need refurbishment
- § BLY Canada and Laos are trying to move those LF-90 from Indonesia

BOART LONG YEAR LF90

Buyback clausal Options - 01



- Ormat to buy refurbished rig from Boart Longyear.
- Price USD 400,000
- BL got a buyback right
- Rate:
 - USD 3,434/ shift (drilling)
 - USD 2,434/ shift (any non-drilling shift or standby)

Rig Owned by ORMAT Options - 02



- ORMAT to buy refurbished rig from Boart Longyear.
- Price USD 400,000
- No buyback right
- Rate:
 - USD 3,434/ shift (drilling)
 - USD 2,434/ shift (any non-drilling shift or standby)

Rig Owned by Boart Longyear Options - 03



- All refurbished will be sponsored by ORMAT
- Investment USD 175,000 – 200,000
- Deposit USD 350,000
- Upfront Mobilization cost
- No buyback right
- Rate:
 - USD 3,500/ shift
 - USD 2,500/ shift

EPIROC – C5

- § EPIROC is not able to provide routine maintenance
- § EPIROC is not able to run the rig

1.1.2 Financial Offer & Lead Time

A. PRICE

No	Description	Qty	Price (USD)
1	Boyles C5 Drilling Rig	1	341,325

TERMS OF PAYMENT

- All clauses will be based on PT Epiroc Southern Asia Standard Terms and Conditions
- Currency: USD (currency is in USD and will be converted to IDR based on the exchange rate when PO received by Epiroc)
- Incoterms: DDP Jakarta
- Exclude: VAT 11%

COMMERCIAL TERMS & CONDITIONS AND WARRANTY

1.1.3 Validity of Proposal

- Lead time Boyles C5: 12-14 Weeks Ex-works Nanjing-China (after PO), China factory + 4 weeks shipping time to Indonesia
- Quotation validity : 7 days

1.1.4 Payment Terms and cancellation

Payment Milestone	Value (%)
Advance against Purchase Order	30%
Payment at Ex Works Readiness	50%
Payment at Delivery Point (According to Delivery Terms)	20%
Cancellation Fee Milestone	Value (%)
After Order Date	5%
180 Days Prior to Completion	15%
90 Days Prior to Completion	25%
Rig Completed	30%

EPIROC – C6

- § EPIROC is not able to provide routine maintenance
- § EPIROC is not able to run the rig

1.1.2 Financial Offer & Lead Time

A. PRICE

No	Description	Qty	Price (USD)
1	Boyles C6 Drilling Rig	1	408,210

TERMS OF PAYMENT

- All clauses will be based on PT Epiroc Southern Asia Standard Terms and Conditions
- Currency: USD (currency is in USD and will be converted to IDR based on the exchange rate when PO received by Epiroc)
- Incoterms: DDP Jakarta
- Exclude: VAT 11%

COMMERCIAL TERMS & CONDITIONS AND WARRANTY

1.1.3 Validity of Proposal

- Lead time Boyles C6: 14-16 Weeks Ex-works Nanjing-China (after PO), China factory + 4 weeks shipping time to Indonesia
- Quotation validity : 7 days

1.1.4 Payment Terms and cancellation

Payment Milestone	Value (%)
Advance against Purchase Order	30%
Payment at Ex Works Readiness	50%
Payment at Delivery Point (According to Delivery Terms)	20%
Cancellation Fee Milestone	Value (%)
After Order Date	5%
180 Days Prior to Completion	15%
90 Days Prior to Completion	25%
Rig Completed	30%

Section	Estimated Cost (USD)	No of Days	Total (USD)	Remarks
New Rig	400,000	-	400,000	Include mob cost
Additional Equipment	1,000,000	-	1,000,000	

CSD 1,800 –
GLOBAL DRILL

CONCLUSION



CONCLUSIONS:

It is feasible to deploy by helicopter, a drilling rig, capable of meeting the base scope requirements of this study.

Based on research to date, only refurbished rigs are available to achieve this within current market constraint.

Based on research to date, no API compliance BOP system, that fits within the core rig envelop was available.

Further research required into ancillary and downhole tool requirements.

Each potential well site will exhibit it's own unique constraints and opportunities. These will need detail assessment on case-by-case basis



FOLLOW-UP RESEARCH

FOLLOW-UP RESEARCH:



Source globally alternate vendors for purchase and rental options.



Further investigation required into helicopter availability, constraints and operations.



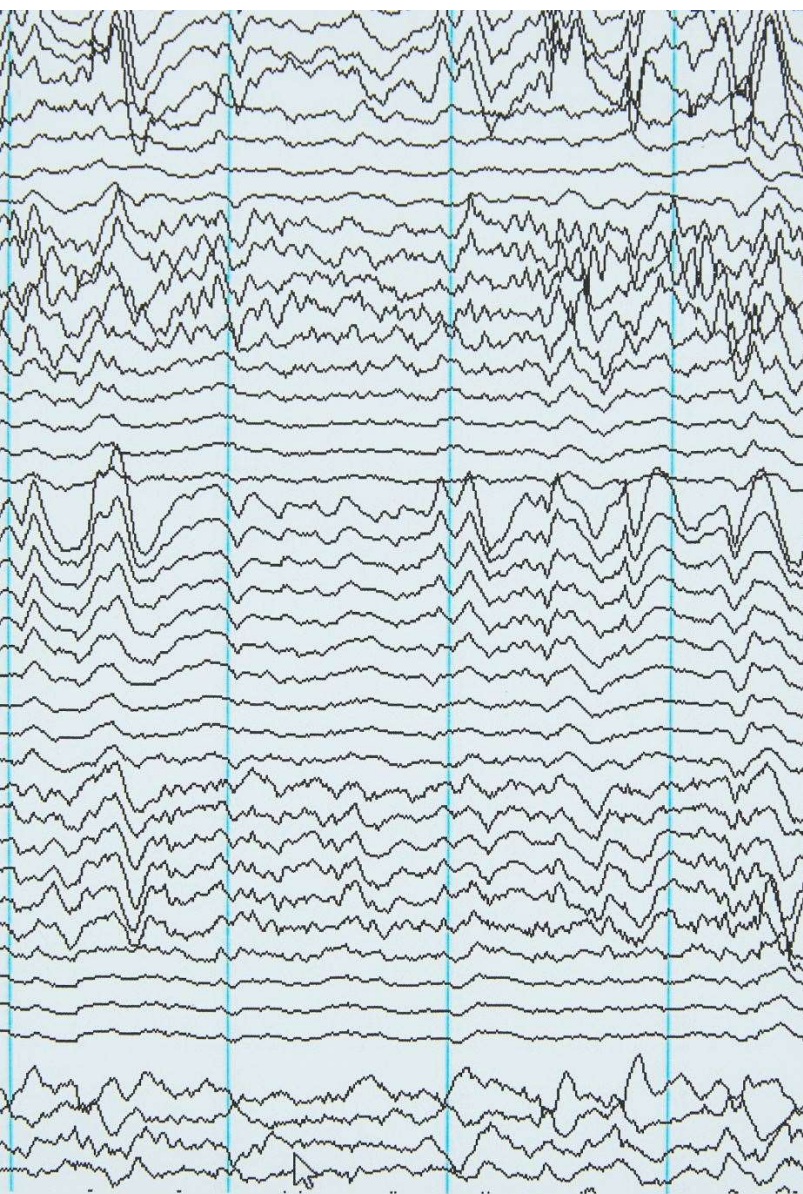
Detail investigation to be carried out, using actual prospective site location, to give more robust cost estimations.



To comply with normal drilling safe practice, further investigation is required to find alternate API compliant well control systems.



Ormat resources to review existing proposals and develop proprietary well design.



APPENDIXES — TECHNICAL DATA

BOART LONGYER

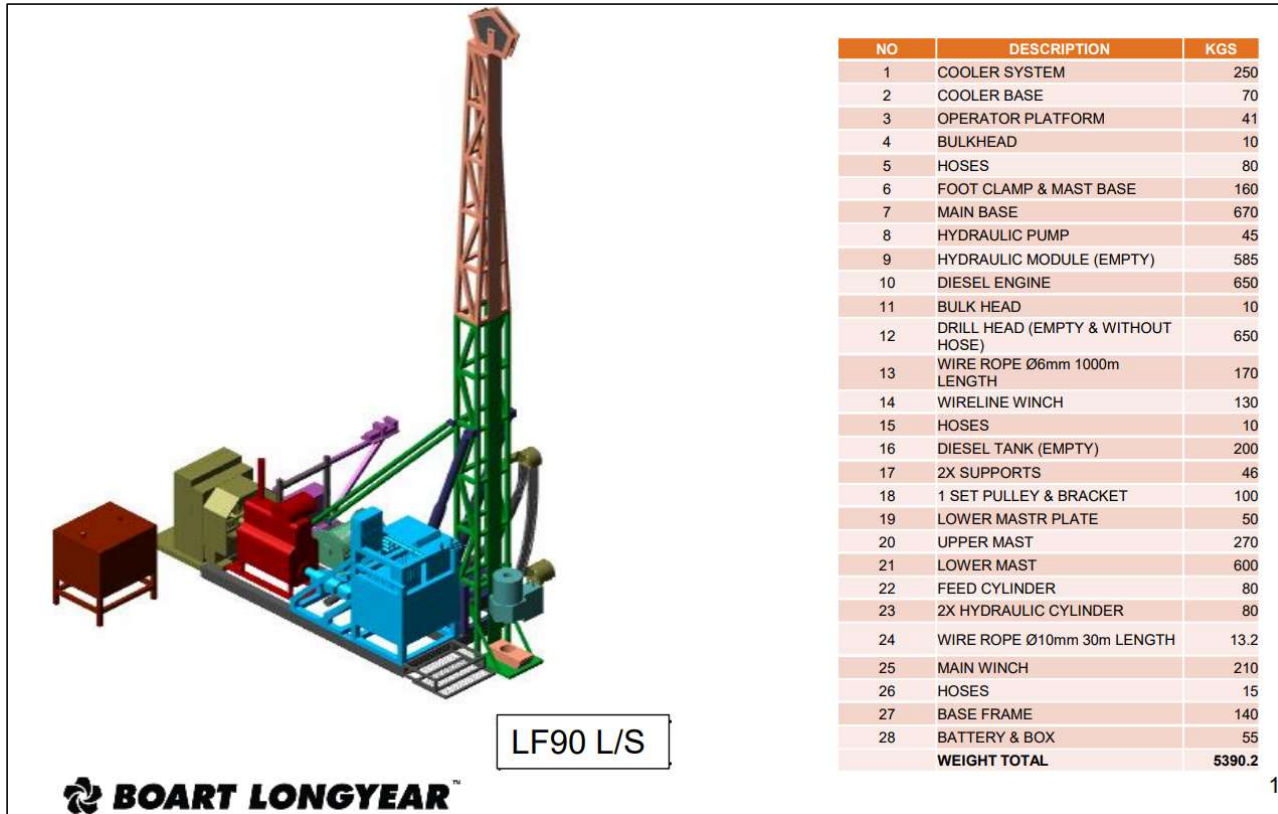
OPTION#01



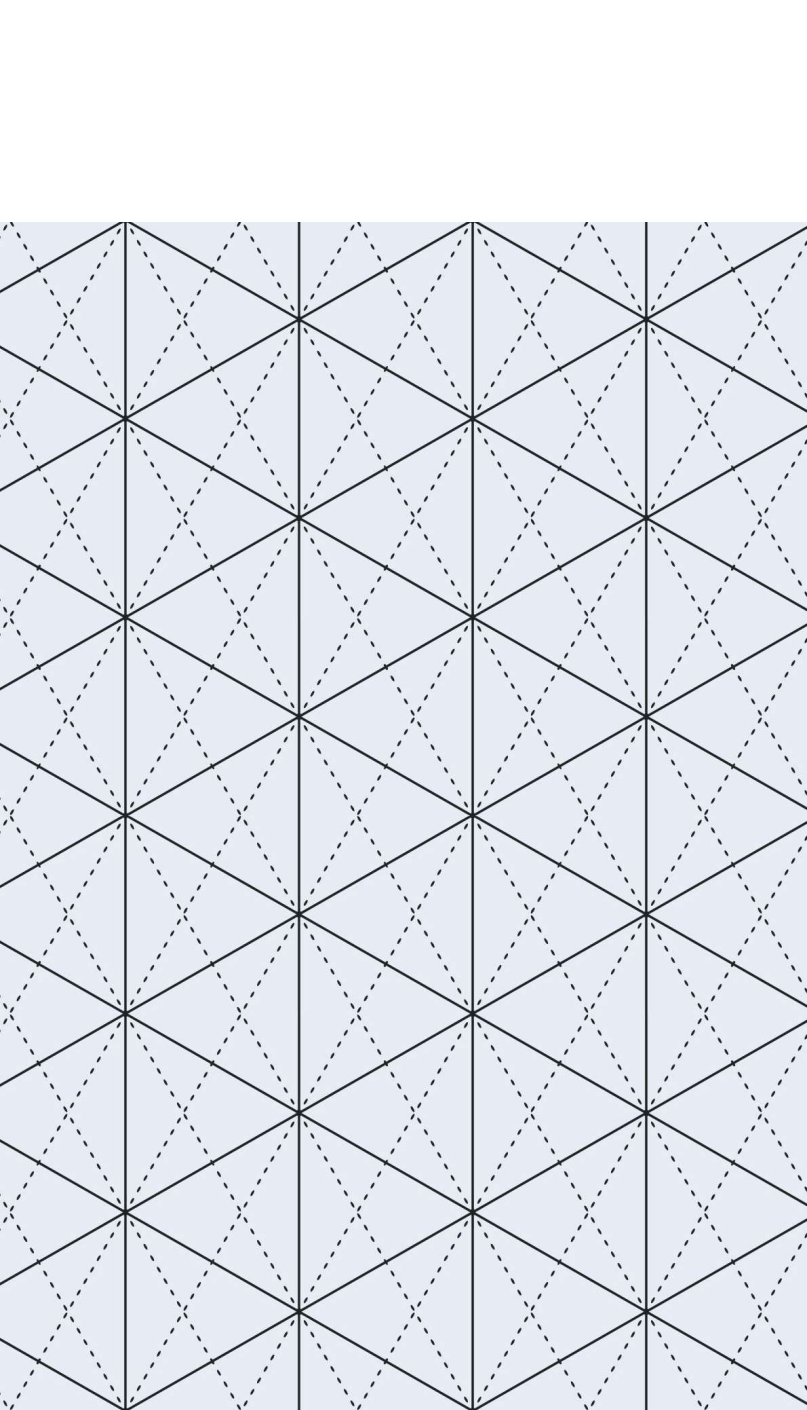
- Total Weight: 5.4 Ton
- Maximum Weigh is 670 kg
- No of coring rig loads: 12-14 load
- Auxiliary equipment's load: up to 16 loads
- LF-90 have been discontinued
- Typical well pad sizes is 20 m x 20 m to 30 m to 30 m

Drilling Depth Guidelines				
Drill Rod / Core Barrel	Dry Hole		Fluid Filled	
	Hole Depth (Meters)	Hole Depth (Feet)	Hole Depth (Meters)	Hole Depth (Feet)
BRQ / BQ	1 210	3,698	1 385	4,543
BRQTK / BQTK	1 523	4,995	1 744	5,722
NRQ / NQ / NQ2	930	3,053	1 064	3,492
NRQ V-WALL™	1 052	3,451	1 193	3,915
HRQ / HQ	631	2,071	722	2,370
HRQ V-WALL™	795	2,607	898	2,947
PHD / PQ	417	1,368	476	1,562
PHD V-WALL™	568	1,865	637	2,089

LF-90



LF-90 LOAD BREAKDOWN



LF-90

Below are some facts for LF-90 that received from Boart Longyear:

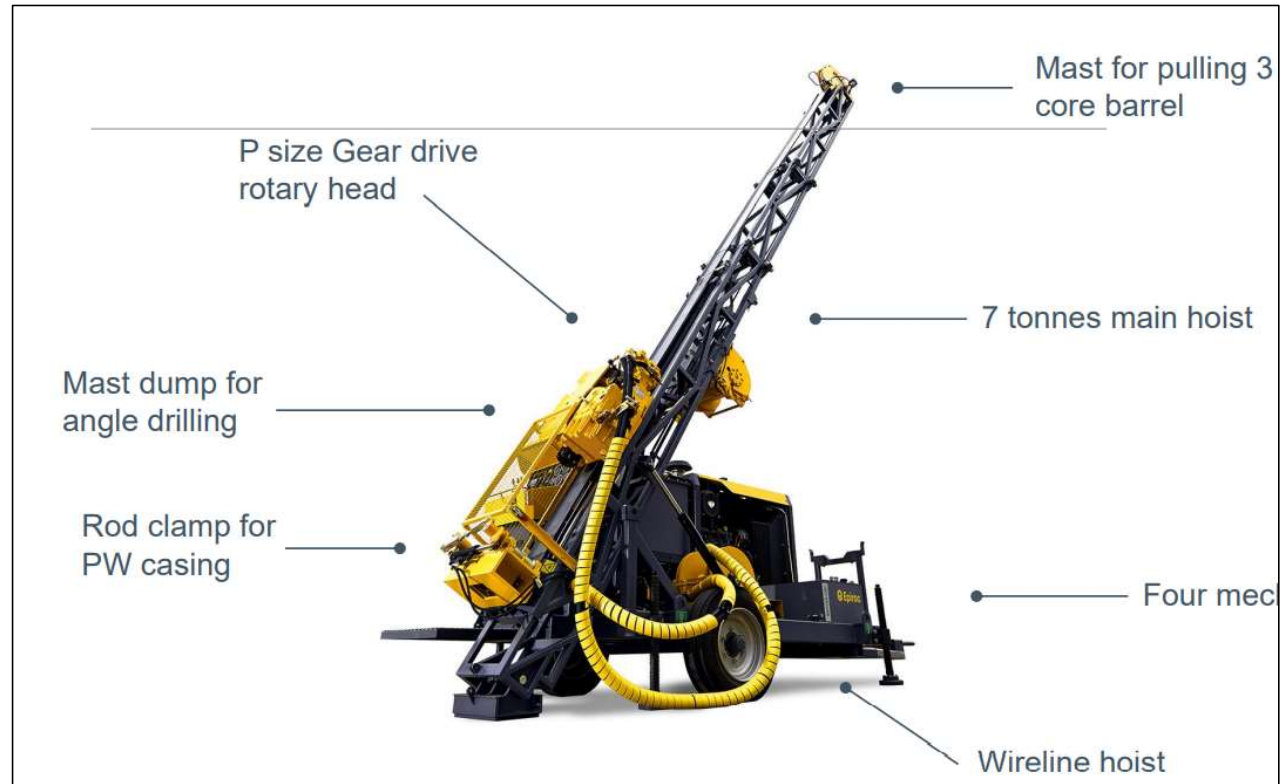
- There are two LF-90 in Indonesia that need to be refurbished prior to the utilization
- LF-90 manufacturing process have been discontinued globally, but issues are not expected on the spare part availability from Boart Longyear.
- Boart Longyear Canada and Laos are looking to occupied both of LF-90 that are currently available in Indonesia.

EPIROC

OPTION#02

BOYLE C-5 LOAD DETAIL

- Maximum components load 900 kg
- Tailored lifting point for Helicopter transport
- Total weight – 5,000 kg
- Rig will be broken into 7 main modules





	Module	Weight(Kg)	Further break down	Weight (Kg)
1	Powerpack	900	Powerpack	900
2	Hydraulic Tank	645	Triple pump	95
			Tank structure	230
			Hydraulic Oil	320
3	Base	985	Fuel tank	55
			Chassis	630
			Jack kit	100
			Tyre kit	140
			Battery box	60
4	uper&middle mast	520	upper&middle mast	520
5	Rotation Unit	655	Rotation Unit	655
6	Lower mast	980	lower mast	860
			Telescoping mast	120
7	Others	1050	Main winch	290
			Rod clamp	170
			Towbar	100
			Wireline winch	175
			operation platform	25
			Raise cylinder (2X)	45
8	Option-Mud pump	320	Control panel	245
			Mud pump	320

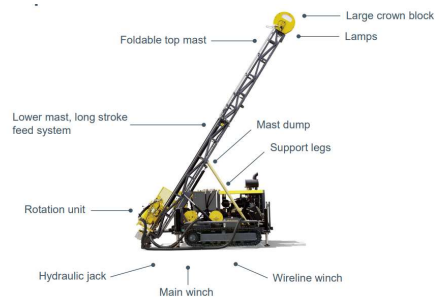
BOYLE C-5 LOAD DETAIL

BOYLE- C-5

Below are some facts for Boyle – C-5 that received from EPIROC:

- EPIROC proposed the Boyle C-5 coring rig, manufacturing process will be done in China with 4-5 months of lead time

BOYLE C-6 LOAD DETAIL

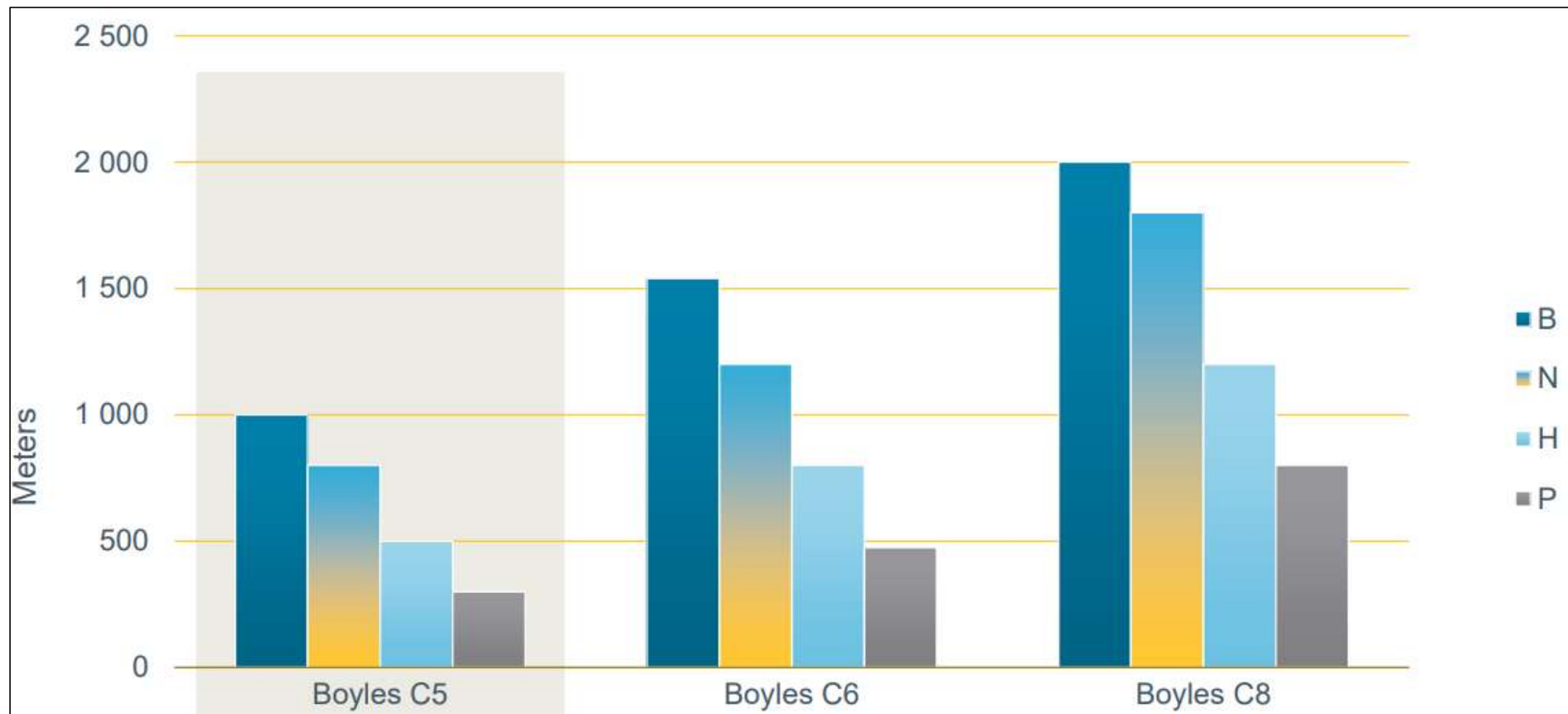


- Deeper coring capability – 1,200 m N-size
- Maximum components load 1,500 kg
- Tailored lifting point for Helicopter transport
- Total weight – 7,000 kg. Total 9,000 kg with Crawler
- Equipped with Radio Remote control



Boyles C6		component weight (kg)
Lower mast(Incl. Rotation unit)		1515
Top mast		420
Rod holder		220
Support leg(2X)		80
Safe Guard		35
Main hoist		290
Wireline winch		175
Subframe for winches and mast dump		250
Hydr.tank		235
Hydr.oil cooler		68
Power unit		1100
canopy		780
tyre kit		280
wheel shaft		120
trailer frame		100
water pump		230
control panel		220
fuel tank		60
Tow bar		101
wagon frame		1100
total		7379

BOYLE C-6 LOAD DETAIL



CORING DEPTH CAPACITY

PT CIME

OPTION#03

CD – 2,000

- Current Location in Australia
- Total 16 rig loads
- Heaviest Component @ 750 kg
- Total weight is 9,2 Tons with crawler

ROD	Depth (m)
PQ	700
HQ	1,000
NQ	1,550



CD – 2,000

Major drilling is not selling a rig

Small BOP schematic and pictures

850 kg helicopter

17 loads

Additional loads

Possibility of integrated services

3 months for re-build the rig

45 days coring for 800 m. total 3-4 months

Weight restriction in 800 kg

GLOBAL DRILLING

OPTION#04

AD - 1,000

- Total Weight @ 4,2 Ton
- Total 12-15 rig loads
- Heaviest Component @ 595 kg
- Currently on job at Valley – Sumbawa since 2016
- Ready in Indonesia – 7 units of AD 1,000



Cellar Dimension
0.5 m x 0.5 m x 0.5 m



ROD	Depth (m)
-----	-----------

PQ	400
----	-----

HQ	700
----	-----

NQ	1,000
----	-------

SCD1800H



- Total Weight @ 9,2 Ton
- Total 16 rig loads
- Heaviest Component @ 750 kg
- Currently on job at Valley – Sumbawa since 2016

ROD	Depth (m)
PQ	700
HQ	1,200
NQ	1,700

No.	Description	Weight
1	Upper Mast	485kg
2	Mid Mast	320kg
3	Lower Mast	720kg
4	Foot Clamps	230kg
5	Winch & A Frame	750kg
6	Engine 1	705kg
7	Engine 1	705kg
8	Hydraulic Tank & Pump	675kg
9	Control Panel	440kg
10	Mud Pump & Base	685kg
11	Two Engine Base Module	685kg
12	Mud Pump	770kg
13	Battery Box	295kg
14	Radiator 1	295kg
15	Radiator 2	275kg
16	Rotation Unit	660kg
15	Wireline Drum	660kg
16	Guards	220kg

SCD1800H – LOAD LIST

ID2000

- Total Weight @ 9,6 Ton
- Total 20 rig loads
- Heaviest Component @ 705 kg

ROD	Depth (m)
PQ	800
HQ	1,500
NQ	2,000

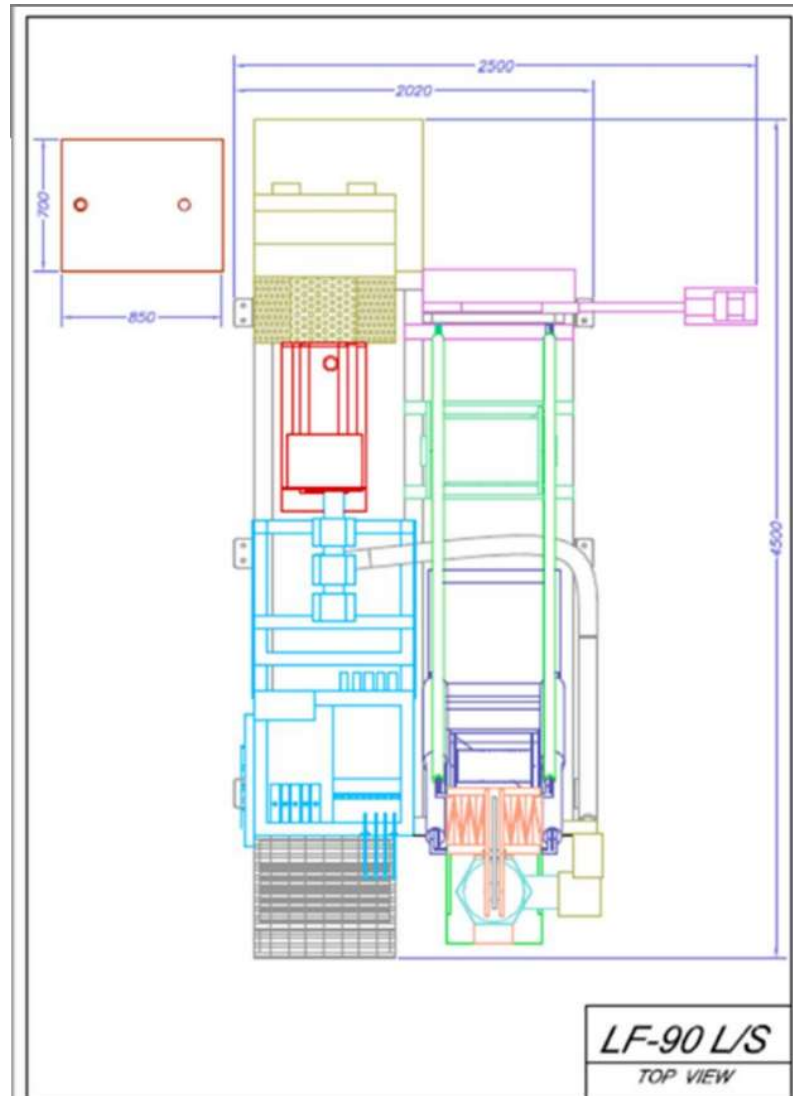


COMP. No.	COMPONENT	WEIGHT (KG.)
1	Cummins Engine 1	705
2	Cummins Engine 2	705
3	Upper Mast Assembly	685
4	Lower Mast Assembly	680
5	Mast Base Plate with Swing Arm, Breakout & Hydraulic Clamp	550
6	Feed Ram	180
7	Mast Supporter & Lift Cylinder	520
8	Mast Dump Frame	300
9	Mast Back Bracket	120
10	Hydraulic Tank with Hydraulic Oil	675
11	2 x Hydraulic Pumps with Hydraulic Hoses	250
12	Wireline Winch with Cable	640
13	Main Winch	660
14	Control Panel with Hydraulic Hoses	640
15	2 x Mud Pumps with Brackets	650
16	Skid Base Frame with Legs (1)	415
17	Skid Base Frame with Legs (2)	400
18	Rotation Head Unit	555
19	Water Tank, Radiator 1 & Fuel Tank	295
20	Water Tank, Radiator 2 & Battery Box	275
TOTAL		9,620

ID 2,000



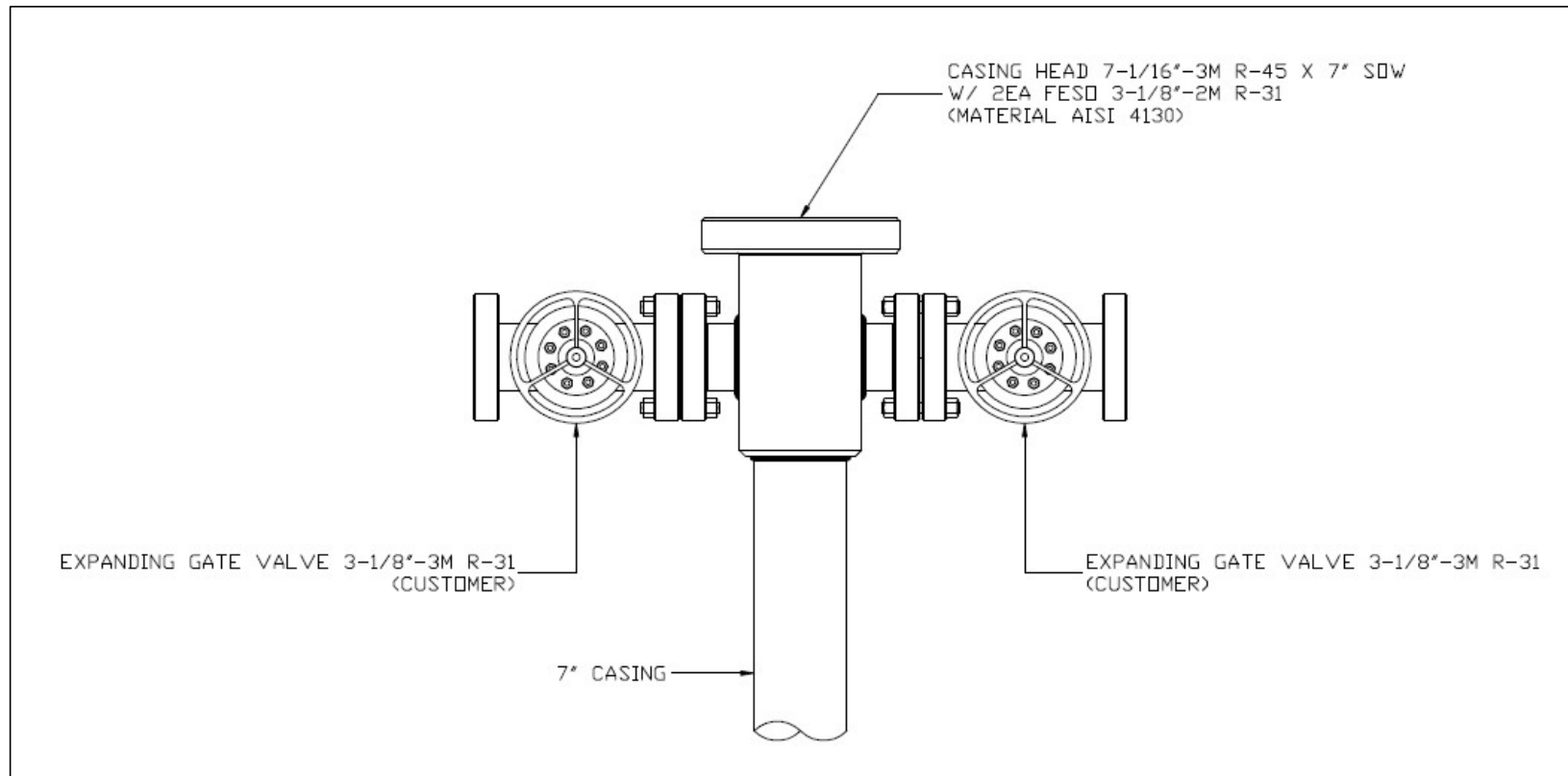
WELL PAD LAYOUT



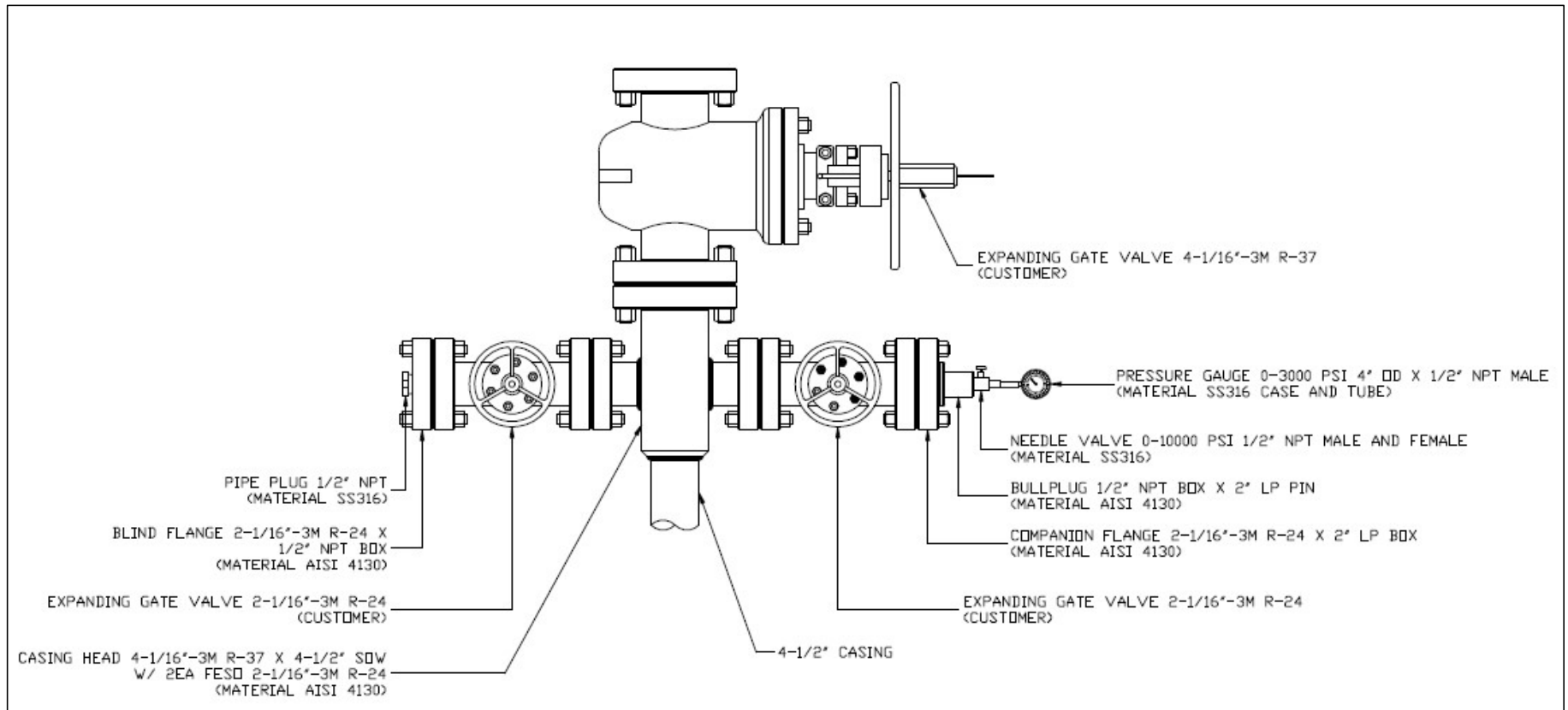
LF-90 – BOART LONGYEAR

WELLHEAD AND MASTER VALVE

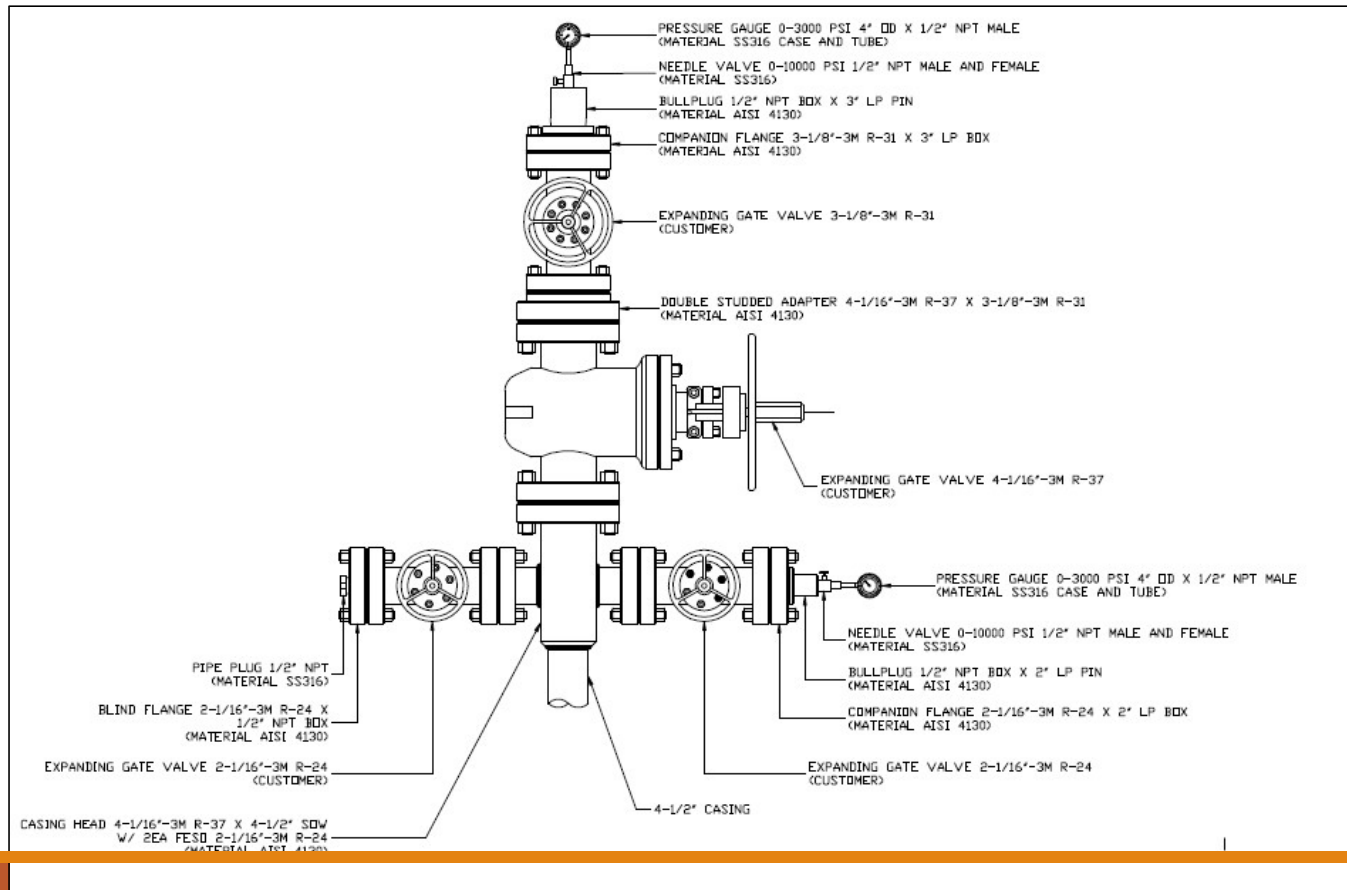




7" STARTING HEAD - SOW



4-1/16" PERMANENT WELLHEAD AND MASTER VALVE

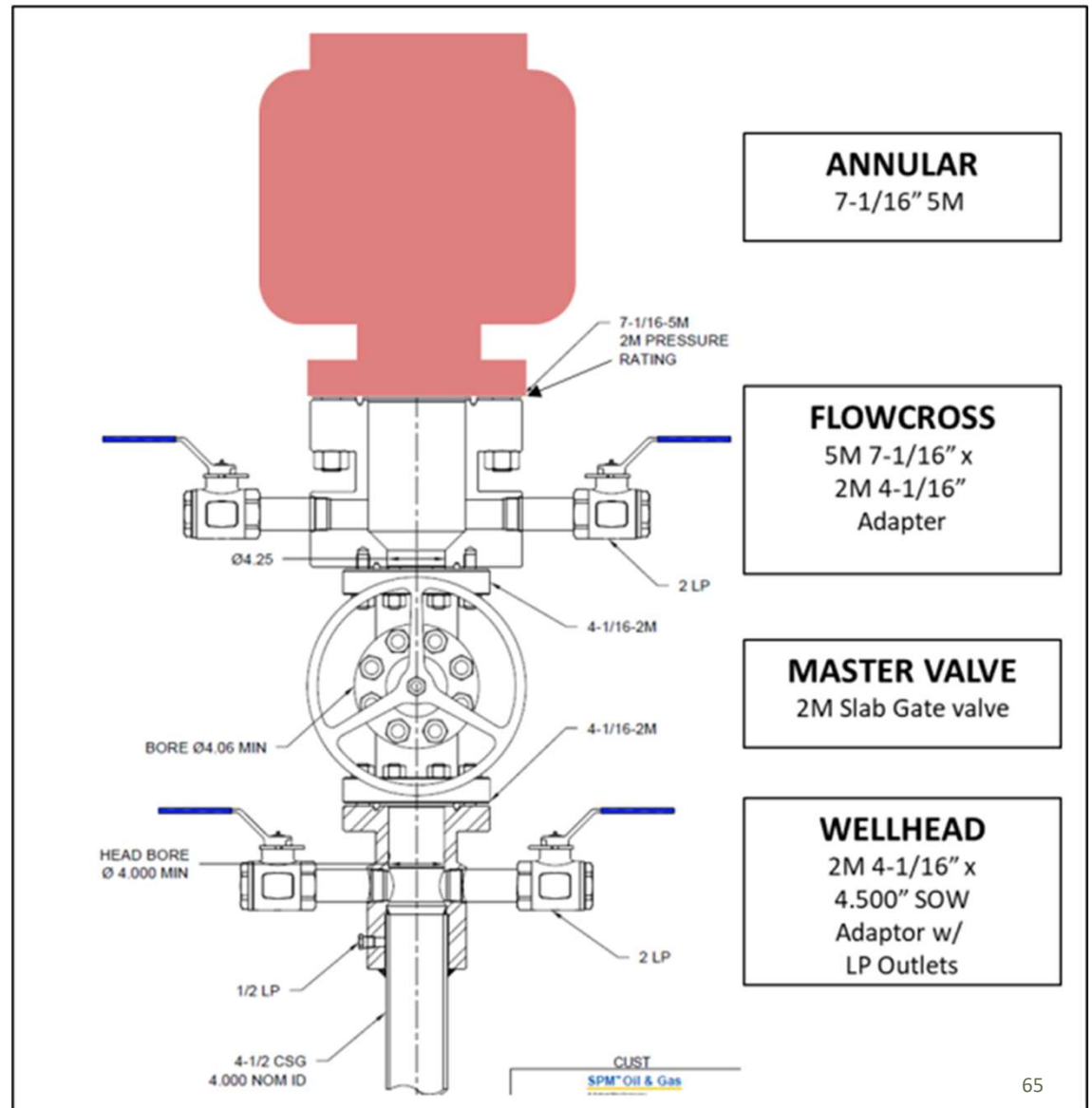


4-1/16" PERMANENT WELLHEAD AND MASTER VALVE – COMPLETION

BOP SYSTEM

CURRENT BOP SCHEMATIC UTILIZED AT CRESCENT VALLEY

- Having 4-1/2" Casing welded to 4-1/16" permanent wellhead, allow the surface BOP connection as per API standard.
- BOP system may be able to be adopted in Indonesia, by having the helicopter coring rig, that allow to accommodate the height of BOP components.
- In addition, BOP components dimension and weight need to be inside the Helicopter transportation limitation.



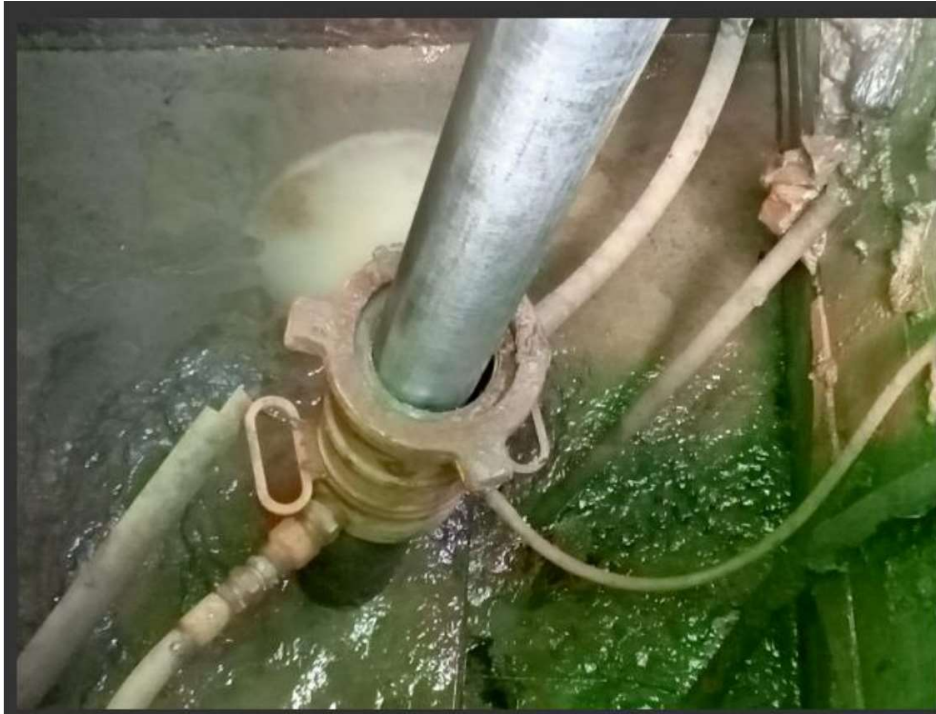
ANNULAR BAG

Below Annular bag is utilized by Boart Longyear and GlobalDrill in Helicopter Coring rig operation in Indonesia.

Potential Issues:

- No- API
- Diverting line ID may be too small to divert the flow from the well





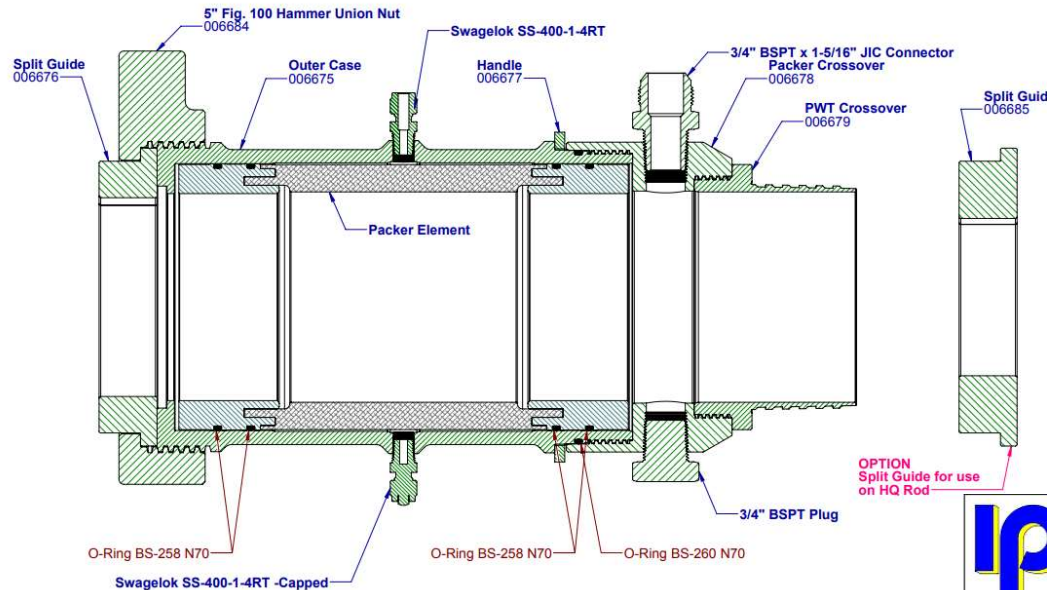
Annular Diverter being attached to PWT casing and drilling using PQ size to planned depth.



IPI (INFLATABLE PACKER INTERNATIONAL) ANNULAR DIVERTER

IPI (INFLATABLE PACKER INTERNATIONAL) ANNULAR DIVERTER

- Equipped with two side ports for kill line and bleed offline
- Diverting the influx / flow from annulus to sump pit
- Kill line to pump water trough annulus to kill the well
- Packer material is HNBR – Hydrogenated Nitrile Rubber – 25 – 160 deg C

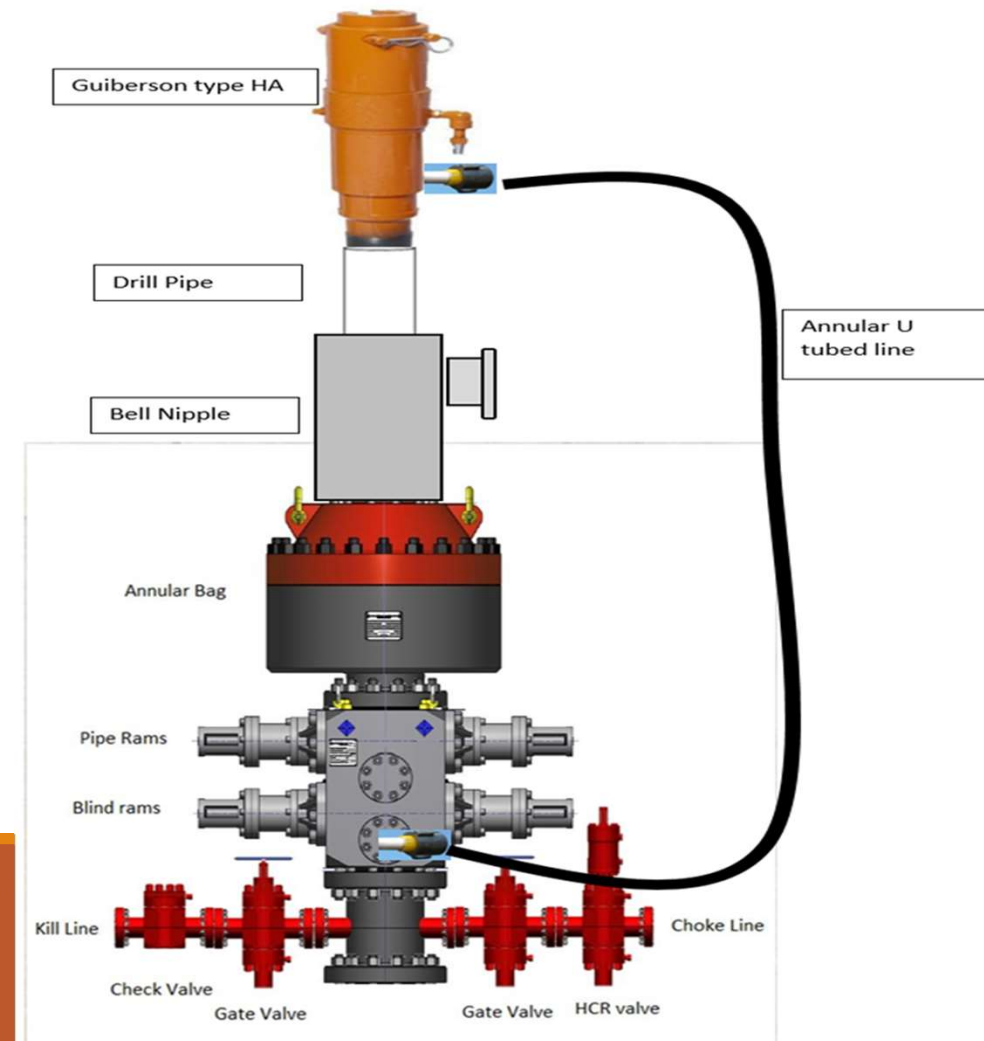


IPI Inflatable Packers
International Pty Ltd
1 Pitino Court, Osborne Park, Western Australia 6017
Phone: +618 9204 2448, Fax: +618 9204 2449
www.inflatable-packers.com
info@inflatable-packers.com © 2016

MEASUREMENTS IN MILLIMETERS UNLESS STATED

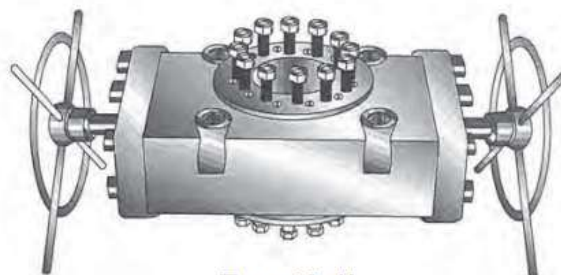
DRN: CM	PQ-PWT Annular BOP		
DRN DATE: 21/07/16	Wednesday, 24 August 2016 11:01:16 AM		
REVISION # 01	APRD: RJM	DRW No. A-10383	
REV DATE: 24/08/16	AprvDATE: 29/07/16	SCALE: 1:2	
ISSUED FOR: INFORMATION ONLY			

Hydraulic Wireline Oil Savers



WIRELINER STRIPPER- GUIBERSON

NOMINAL FLANGE SIZE	Type 'A-1' Manual Drive	Type 'A-1' Chain Drive	Type 'H-1' Dual Hydraulic
	6 in.		
FLANGE SERIES	400 - 600 600 - 900	400 - 600 600 - 900	400 - 600 600 - 900
CASING SIZE	5 - 7 in.	5 - 7 in.	5 - 7 in.
APPROXIMATE WEIGHT	1500 lbs / 680 kg	1630 lbs / 740 kg	3400 lbs / 1540 kg
OVERALL LENGTH, RAMS CLOSED	48 in / 1219.20 mm	48 in / 1219.20 mm	80-1/4 in / 2038.35 mm
OVERALL HEIGHT LESS STUDS AND HANDWHEEL	11-1/2 in. / 292.10 mm	11-1/2 in. / 292.10 mm	24 in. / 609.60 mm
OVERALL WIDTH LESS HANDWHEEL	24-1/2 in. / 622.30 mm	27 in. / 685.80 mm	24-1/2 in. / 622.30 mm
OPENING THROUGH PREVENTER	7-3/8 in. / 187.33 mm	7-3/8 in. / 187.33 mm	7-3/8 in. / 187.33 mm
WORKING PRESSURE*	3000 psi / 20700 kPa	3000 psi / 20700 kPa	3000 psi / 20700 kPa
TEST PRESSURE	6000 psi / 41400 kPa	6000 psi / 41400 kPa	6000 psi / 41400 kPa
HANDWHEEL DIAMETER	32 in. / 7212.8 mm	32 in. / 7212.8 mm	32 in. / 7212.8 mm



**Type 'A-1'
Manual BOP**



Type 'G' Blowout Preventer

Table 48.1: TYPE 'G' BOP - ENGINEERING DATA

TUBING O.D.	2-3/8 in.	2-7/8 in.	3-1/2 in.
Working Pressure	1500 psi / 10300 kPa	1500 psi / 10300 kPa	1500 psi / 10300 kPa
Test Pressure	3000 psi / 20700 kPa	3000 psi / 20700 kPa	3000 psi / 20700 kPa
Approximate Weight	30 lb / 14 kg	40 lb / 18 kg	50 lb / 23 kg
Overall Height	9-1/4 in. / 234.9 mm	9-3/4 in. / 247.65 mm	10-1/4 in. / 260.35 mm
Overall Width	3-1/4 in. / 82.55 mm	3-3/4 in. / 95.25 mm	4-1/4 in. / 107.95
OVERALL LENGTH, RAMS CLOSED	18-1/2 in. / 469.90 mm	18-1/2 in. / 469.90 mm	20 in. / 508 mm
TUBING THREAD SIZE	EUE 8RD / 499.90 mm	EUE 8RD / 499.90 mm	LP 8V / 508 mm

DIVERTER SYSTEM - GUIBERSON

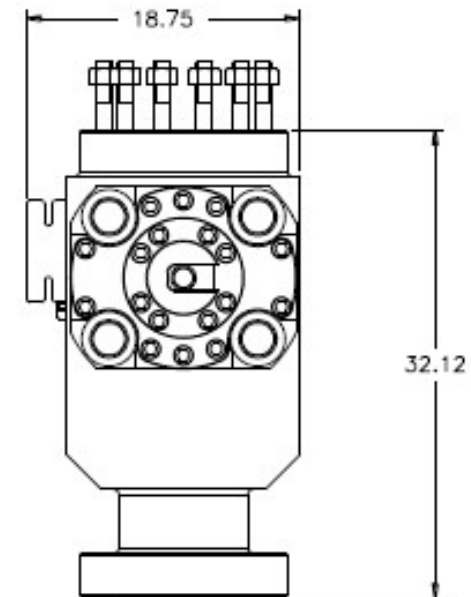
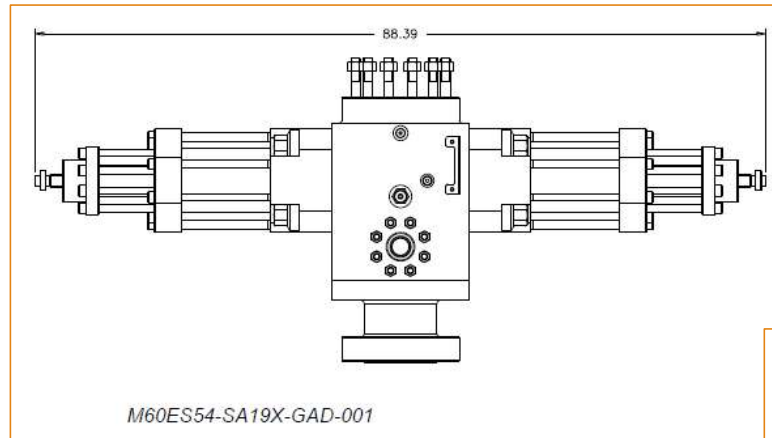
NOV – TEXAS OIL TOOLS 4-1/16" ES SINGLE RAM

Descriptions:

- Designed for coil tubing and snubbing ops
- Country of origin – US
- Working pressure 10,000 psi
- Hydrostatic and F/T as per API 6A
- Dimension: Height: 0.816 m and length: 2.3 m

Potential Issues:

- Weight 2,500 lbs
- Handling issue by the coring rig during operation
- Exceed the helicopter weight limit

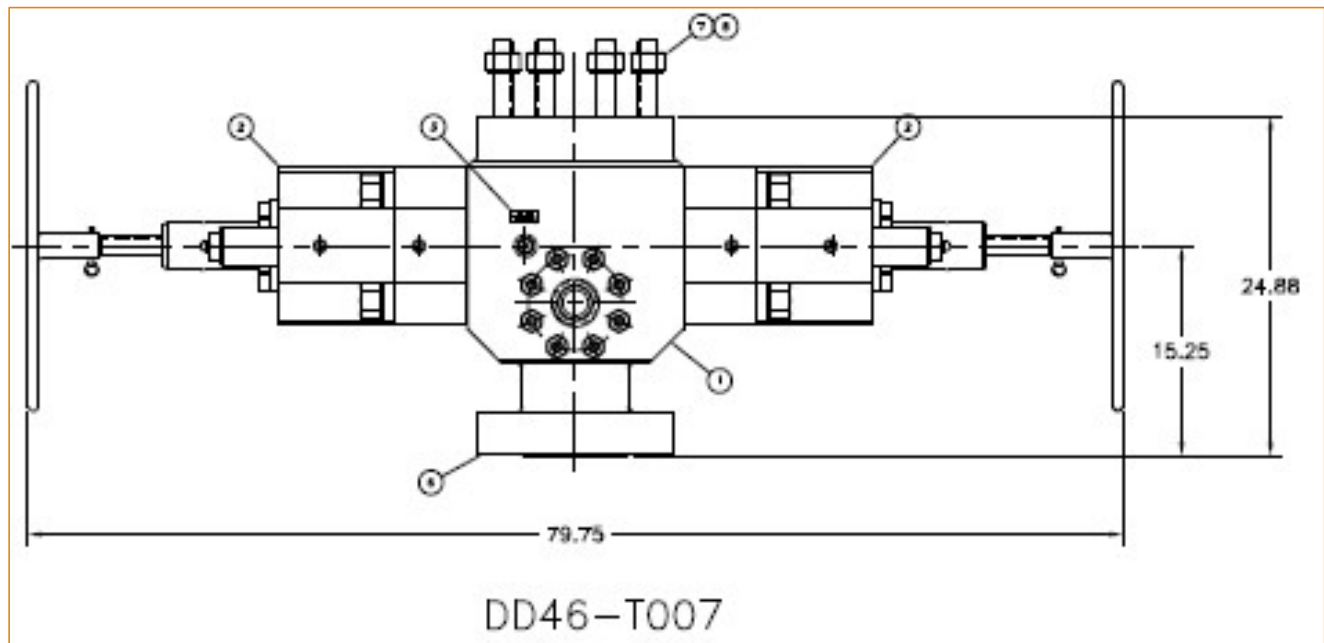


Descriptions:

- Designed for snubbing ops
- Country of origin – US
- Working pressure 15,000 psi
- Hydrostatic and F/T as per API 6A
- Dimension: Height: 0.718 m and length: 2.0 m

Potential Issues:

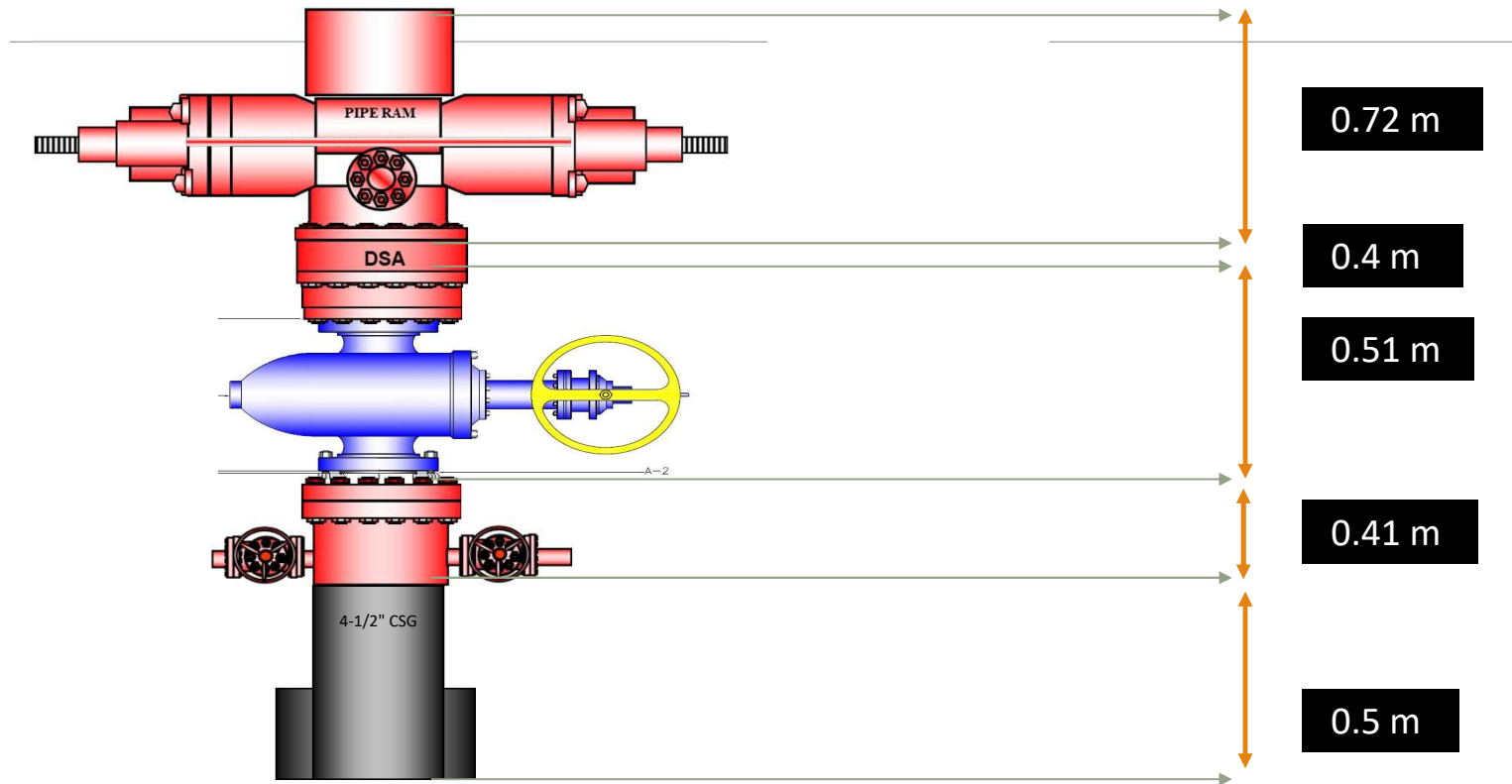
- Weight 1,966 lbs
- Handling issue by the coring rig during operation
- Exceed the helicopter weight limit



WEATHERFORD – TEXAS OIL TOOLS 4-1/16" DD46 SNUBBING BOP

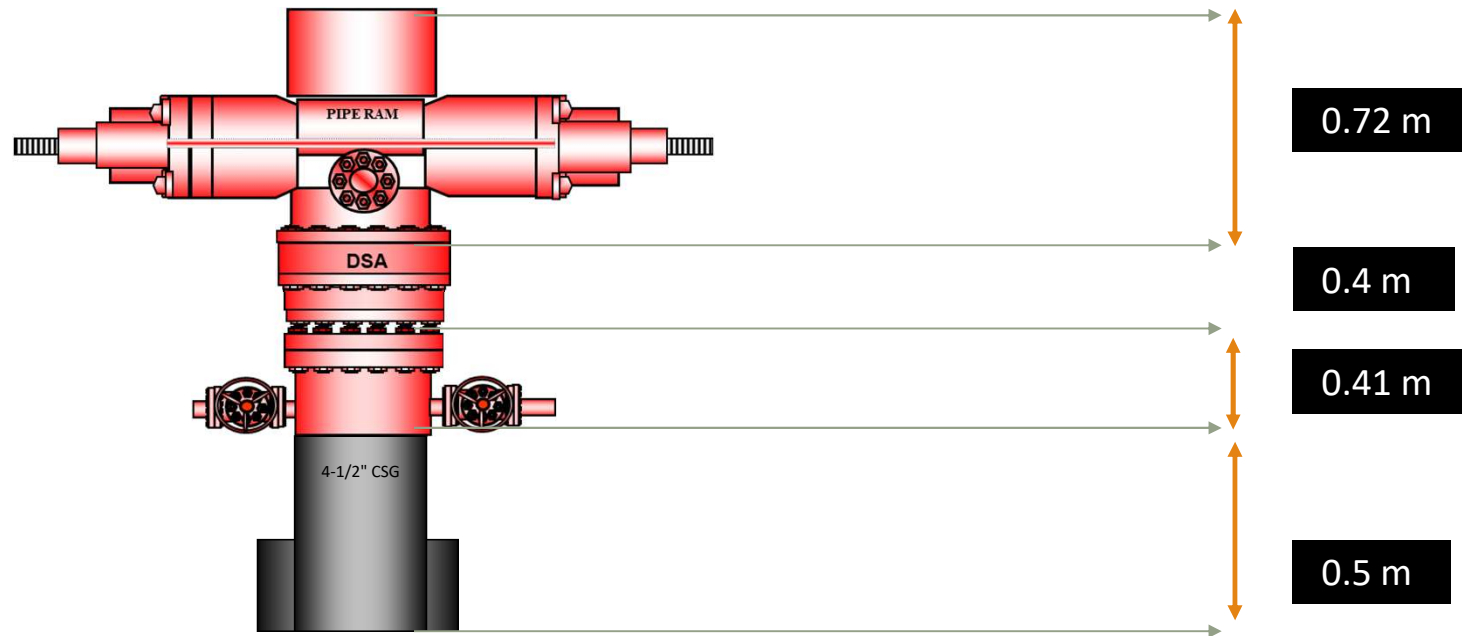
4-1/16" BOP STACK WITH MV

- Casing Head 4-1/16" – 3M – 4-1/2" SOW, R-24 AISI 4130
- Gate valve 4-1/16" – 3 M R-37
- DSAF 4-1/16" 3 M to 4-1/16" 15 M
- Single Ram 4-1/16" 15 M



4-1/16" BOP STACK WITHOUT MV

- Casing Head 4-1/16" – 3M – 4-1/2" SOW. R-24 AISI 4130
- DSAF 4-1/16" 3 M to 4-1/16" 15 M
- Single Ram 4-1/16" 15 M



Blowout Preventers

Bore Size	Pressure	Cameron®	Hydril®	Shaffer®	Other
2" - 3-1/2"					Various manual & hydraulic for wireline
2-9/16"	5M		A		
3-1/16"	15M		SH		
4-1/16"	3M	SH			Regan A
4-1/16"	5M		A		Bowen SM, Dm, Regan A
4-1/16"	10M		A	A, SH	Bowen SM, SH, DH
4-1/16"	15M				Bowen SH, DH, & Foley SH, DH
7-1/16"	2M		A		WSI, KSI, Townsend SM, Regan A
7-1/16"	3M		A	SM, DM, DH, A	Guiberson SM, DM
7-1/16"	5M	SH, DH, A	DH, A	SH, DH, A	Flocan SH, DH, Guiberson DH
7-1/16"	10M	SH, DH, A	SH, DH, A	SH, DH, A	WSI, KSI, Townsend DH
7-1/16"	15M	SH, DH, A	A	SH, DH, A	
7-1/16"	20M	SH, DH, A			
9"	2M		A		
9"	3M		A	SM, DM, SH, DH, A	WSI, KSI, Townsend DH, Regan A
9"	5M		A	DH, A	
11"	3M		A	SM, DM, SH, DH, A	PATCO
11"	5M	SH, DH, A	A	DH, A	
11"	10M	SH, DH, A	A	SM, DM, SH, DH, A	
11"	15M	SH, DH, A	A	SH, DH, A	
13-5/8"	3M		DH, A	SH, DH, A	
13-5/8"	5M	SH, DH, A	SH, DH, A		
13-5/8"	10M	SH, DH, A	A	SM, DM, SH, DH, A	
13-5/8"	15M	SH	SH, DH, A	SH, DH, A	
16-3/4"	2M		A	SH, DH, A	
16-3/4"	3M	SH, DH, A	A	DH	
16-3/4"	5M	SH, A	A	DH	
21-1/4"	2M	SH	A	SH, DH, A	
21-1/4"	3M	SH		SH, DH	
21-1/4"	5M	SH	A	SH, DH, A	
21-1/4"	10M			A	
29-1/2"	500		A		
30"	1M				
30"	3M			A	Presco - SH

KEY: SH - Single Hydraulic SM - Single Manual A - Annular
 DH - Double Hydraulic DM - Double Manual

Note: 562, 675 and 862 Series pumps are also available on request.

CAMERON - SLB

CASE STUDIES

AKUTAN EXPLORATION DRILLING – 2010 - ALASKA

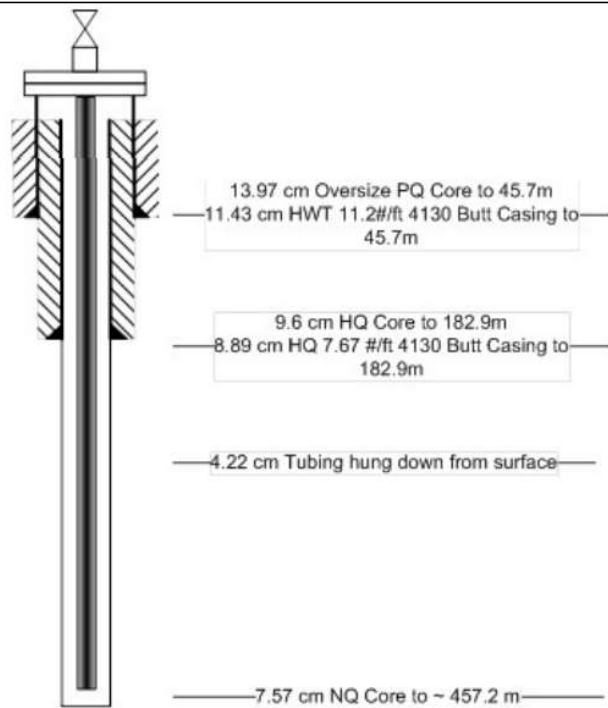


Figure 2. Well design for Hot Springs Well (TG2), with a planned TD of 457.2 m.

- Temperature Gradient data to constrain resource capacity
- Two wells drilled
- Well flow inside core rod during POOH core barrel
- TLC zone at 178 m, MRT temp recorded to 150 C



TOLHUACA – CHILE - 2009

- Targeted to complete the well prior to the winter (extensive snowfall).
- Utilized LF-230 (Boart Longyear) rig
- Rig was helicopter transported to rig site
- Well control event occurs during drilling surface section
- Well control event occurred causing two weeks delay to train the crew for proper well control technique.
- Cellar with 2 m depth was constructed to accommodate low profile annular preventer
- Two helicopter utilized: Kamov with 4,000 kg lifting capacity and Bell 407 with 600 kg lifting capacity at 2,000 m altitude
- Staging area distance to well pad is 6 minutes.
- Well schematic:
 - 5.5" PW casing to 20 m
 - 4.5" HWT to 116 m
 - HQ size cemented at 503 m
 - NQ size to 1,003 m
 - BQ size to 1,073 m



LAGUNA DEL MAULE EXPLORATION DRILLING - 2009

- Operated by Magma Energy Corp
- To prove resources temperature
- Tight timeline
- Resource located 10 km from nearest road. Elevation is 1,200 m
- Vertical depth to 659 m with high downhole temperature @ 200deg C
- Well drilled to 659 m with BQ size of coring rod

CORING ROD

CORING ROD DEPTH LIMITATION

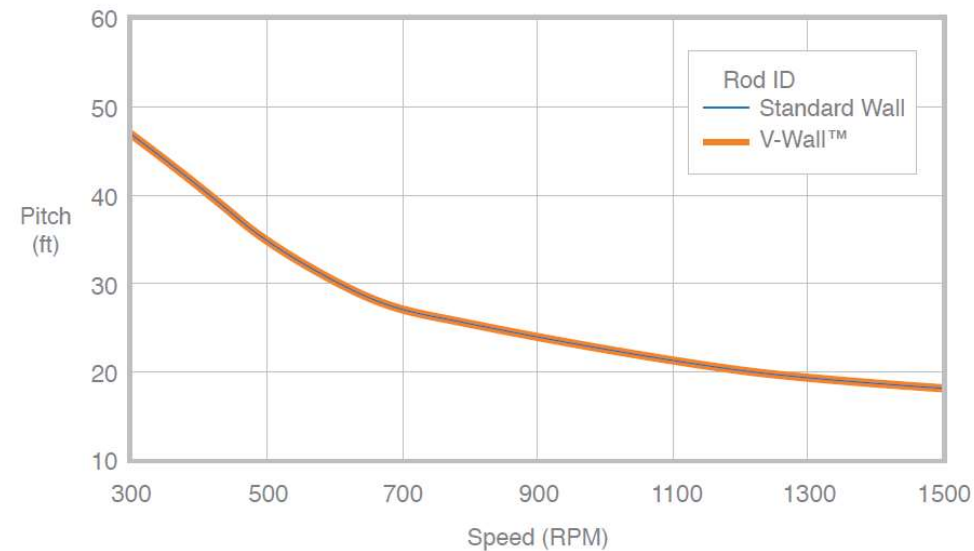
RATED DEPTH CAPACITY

ARQ TK	1500 m	(4,900 ft)
BQ	1500 m	(4,900 ft)
BRQ TK	1500 m	(4,900 ft)
BRQ	3000 m	(9,800 ft)
NQ	1500 m	(4,900 ft)
NQ V-WALL	1680 m	(5,500 ft)
NRQ	3000 m	(9,800 ft)
NRQ V-WALL	3300 m	(11,000 ft)
HQ	1500 m	(4,900 ft)
HQ V-WALL	1680 m	(5,500 ft)
HRQ	2500 m	(8,200 ft)
HRQ V-WALL	3000 m	(9,800 ft)
PHD	1500 m	(4,900 ft)
PHD V-WALL	2000 m	(6,500 ft)

V-WALL advantages:

- 30% less weight
- Decrease on transportation load
- Increase depth capacity
- Faster inner barrel trip because of more clearance

MIDBODY BENDING STRESS



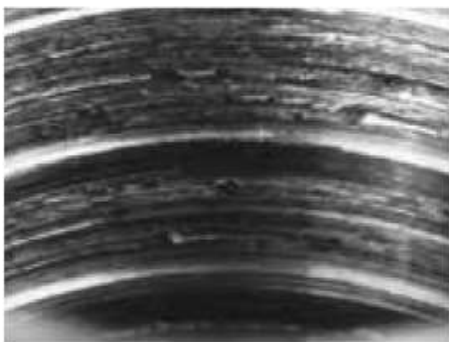
CORING ROD CONNECTION

PIN THREAD WEAR



Boart Longyear™ Patented RQ™ Thread

After 60 make/
break cycles, the
case hardened
pin thread resists
wear.



Competing Thread

After only 30 make/
break cycles, the
non-case-hardened
pin thread allows
adhesion wear.

BOX THREAD WEAR



Boart Longyear™ Patented RQ™ Thread

After 60 make/
break cycles, the
box thread shows
minimal abrasion
wear.



Competing Thread

After only 30 make/
break cycles, the
box thread shows
adhesion wear
which leads to
galling and seized
threads.

SUB-BASE

- Provide space for the BOP
- Height needed: **1 m**
- Structure Engineer – for design
- Certification
- Weight of the sub-base



SUB-BASE

CHALLENGES

CHALLENGES

Limited knowledge and experiences

Limited rig options

Limited contingency string due to small diameter hole

Depth limitation of rig, and coring rod

Kick detection in small annulus

Ability coring rig to handle rotary drilling – flowrate, weight, solid handling, pressure capability